

BiCG-2k

1. Guess x_0 ↑ $AX=b$
Dslash
2. compute $r_0 = b - Ax_0$
3. for $i=0, 1, 2, \dots$; do
4. $e_i \leftarrow \text{solve } Ae_i = r_i$ ↑ Big
5. $x_{i+1} = x_i + e_i$
6. $r_{i+1} = b - Ax_{i+1}$
7. check convergence and break;
8. end

$$\underline{x_c} = x_0 + e_c$$

AMG, BiCGstab-2k

1. Guess x_0
2. compute $r_0 = b - Ax_0$
3. for $i=0, 1, 2, \dots$; do
4. $e_i \leftarrow \text{solve } Ae_i = r_i$
5. $x_{i+1} = x_i + e_i$ → BiCG 2步法/精
6. $r_{i+1} = b - Ax_{i+1}$
7. $r_{i+1}^c = R r_{i+1}$ ↑ 粗 ↑ 粗 ↑ 粗
8. $e_{i+1}^c \leftarrow A^c e_{i+1}^c = r_{i+1}^c$, On Coarse Grid ← $e_{i+1} = P e_{i+1}^c$
9. $x_{i+1} = x_{i+1} + P e_{i+1}^c$
10. $r_{i+1} = b - Ax_{i+1}$
11. check convergence
12. end

smoother

pre-sm

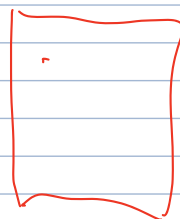
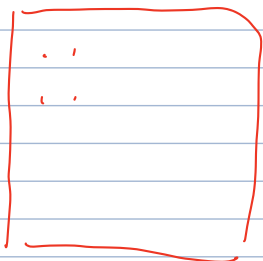
Restriction

coarse grid

Prolongation

后光滑

$$R = P^H \leftarrow \text{Vect}[12/14/16] \leftarrow \text{solve } Ax=0, \text{ BiCG}$$



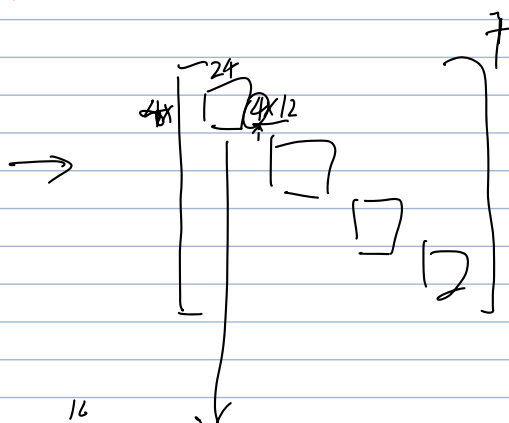
R

test vectors → null space

$$Ax \approx 0 \rightarrow \text{小本征值}$$

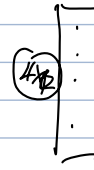
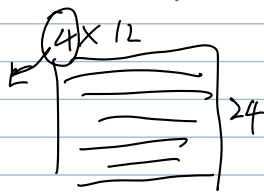
V

$$V_{12} \times 24$$



$$4 \text{ Lat } 12$$

$$(1 \times 24)$$



2x2x2x2